Functions, domain and range, continuity, and graph behavior

Name: _____ Date: _____

A. Is It a Function?

Determine whether the equation defines y as a function of x.

1. $y = -9x^2 + 5x - 2$

2. $x = y^2 + 6$

B. Domain

Find the domain of the given function.

3. $\sqrt{7 - x}$

4. $\frac{x + 9}{(x + 4)(x - 8)}$

C. Range

Find the range of the function.

5. $f(x) = (x - 1)^2 + 4$

D. Continuity

Determine where each function is continuous or discontinuous. If there is a discontinuity, tell whether it is removable or non-removable.

6. $f(x) = \frac{(x - 2)(x + 5)}{x^2 - 4}$

7. $f(x) = \frac{x^2 - 3x}{x}$

E. Analyze a Graph

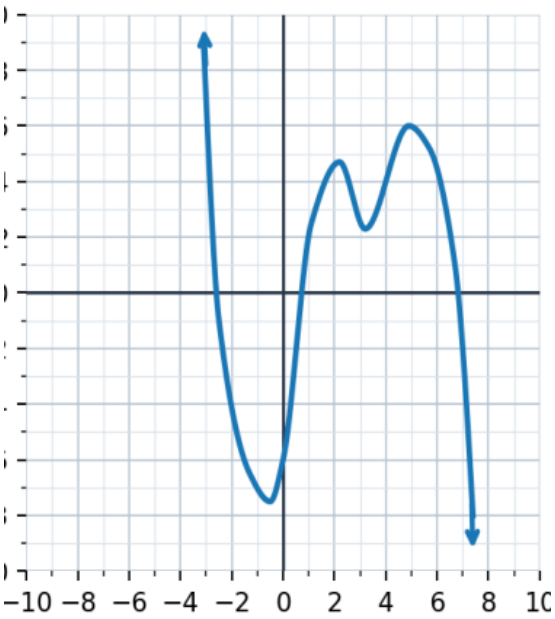
8. Use the graph below to estimate the local maximum and local minimum.

9. Determine the intervals on which the function is increasing, decreasing, and constant.

Graph analysis, boundedness, symmetry, and function operations

Name: _____ Date: _____

E. Graph for Questions 8-9



F. Boundedness

Determine if the function is bounded above, bounded below, bounded or unbounded.

10. $y = x^2 + 5$

11. $\sqrt{9 - x^2}$

12. $y = x^3$

G. Even, Odd, or Neither

Determine whether the function is even, odd, or neither.

13. $y = 3x^2 - 3$

14. $f(x) = 8 + 5x + x^2$

H. Function Operations

Perform the requested operation or operations given f(x) = 7x + 14; g(x) = 4x - 1.

15. $(f + g)(x)$

16. $(f - g)(x)$

17. $(f \cdot g)(x)$

18. $\left(\frac{f}{g}\right)(x)$

Function operations, inverse functions, and transformations

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H. Function Operations (continued)

19. $(f \circ g)(x)$

20. $(g \circ f)(x)$

I. Inverse Functions

Find the inverse of each function.

21. $f(x) = x^3 + 7$

22. $f(x) = \frac{-2x - 7}{6x + 8}$

J. Transformations

Describe how the parent graph $y = x^2$ can be transformed to the graph of the given equation.

23. $y = -2x^2 + 9$

24. $y = (x - 4)^2 + 6$

Chapter 1 Skills Checklist

Check off each skill when you can complete it without notes.

- ☐ Function test; domain and range
- ☐ Continuity and removable discontinuities
- ☐ Local extrema and increasing/decreasing intervals
- ☐ Boundedness and even/odd symmetry
- ☐ Function arithmetic and composition
- ☐ Inverse functions and graph transformations